

POLARIS DC-1 — TECHNICAL SPECIFICATION

SPEC-01: EARTHWORKS AND SUBSTRUCTURE

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Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Synthetic demonstration data. All parties, values and figures in this document are fictional.

1. Scope

- 1.1 This specification covers site clearance, bulk and detailed excavation, controlled blasting, dewatering, made-ground and sulphide-affected material handling, foundations (pad footings, piers and piled/rock-socketed foundations), and the ground-bearing floor slab for the Polaris DC-1 facility, gridlines A–N / 1–24 as defined in PDC-BOD-001 §2.1.
- 1.2 This specification does not cover external drainage, landscaping, road pavements, or the structural steel frame and precast envelope (PDC-SPEC-02), which bear on the substructure described here.
- 1.3 The Contractor's substructure design shall be founded on, and shall not contradict, the ground conditions, bearing values and constraints recorded in PDC-SS-001 Rev B and the foundation strategy set out in PDC-BOD-001 §4.2.

2. Standards and References

Reference	Subject
SFS-EN 1997-1, SFS-EN 1997-2 (Eurocode 7), Finnish National Annexes	Geotechnical design
SFS-EN 1992-1-1 (Eurocode 2), Finnish National Annex	Design of concrete structures
SFS-EN 1990, SFS-EN 1991-1-1/-1-3/-1-4, Finnish National Annexes	Basis of design, actions, snow and wind
SFS-EN 206 + SFS 7022	Concrete — specification, performance, production, conformity (Finnish complement)
SFS-EN 13670 + SFS 5975	Execution of concrete structures
SFS-EN ISO 14688-1/-2	Identification and classification of soil
SFS-EN ISO 22475-1	Sampling and groundwater measurement
RIL 121-2004, RIL 263-2014	Finnish Geotechnical Society guidance — investigation, excavation and support
SFS 4812	Vibration from blasting and piling — guide values near structures
Land Use and Building Act (132/1999) and associated Government Decrees	Statutory basis, blasting notification (louhintailmoitus)

National Building Code of Finland	Structural and fire-safety provisions
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Where a Eurocode part and a cited RIL or SFS guidance document address the same matter, the Eurocode with its Finnish National Annex governs; RIL/SFS guidance is used to interpret local practice within that framework.

3. Materials

Element	Material / class
Granular capping (slab and pavement subgrade)	Crushed rock or approved granular fill, compacted in layers ≤ 300 mm, CBR $\geq 20\%$
Pad and strip footings	Concrete class C30/37, exposure class XC2/XF1 as ground conditions require
Rock-socketed piers, generator and transformer plinths	Concrete class C35/45, exposure class XC2
Ground-bearing slab (white space and E/M support)	Concrete class C32/40, steel-fibre or bar-reinforced per the Contractor's design, exposure class XC1
Reinforcement	SFS-EN 10080, grade B500B
Blinding	Concrete class C16/20, minimum 75 mm
Damp-proof membrane	1200-gauge polyethylene minimum, lapped 300 mm and taped at all joints and penetrations
Movement/isolation joint filler and sealant	Compressible filler with polysulphide or polyurethane sealant, chemical-resistant grade in battery room areas
Backfill to foundations and trenches	Selected site-won till (where free of organics/boulders) or imported granular fill, compacted to 95% max dry density

4. Workmanship and Execution

4.1 Clearance and strip. Topsoil, organic horizons and any made ground shall be stripped from the full building footprint and stockpiled separately. The approximately 900 m² made-ground area in the north-west corner (PDC-SS-001 §7b) shall be excavated and removed, or bridged by a piled slab, only after the archaeological watching-brief condition referenced in PDC-SS-001 §7b has been discharged with the relevant authority.

4.2 Zoned foundation strategy, per PDC-BOD-001 §4.2:

Zone	Gridlines (approx.)	Foundation type
West (Hall 1, primary substation, gatehouse)	1–7	Pad/strip footings on compacted till; no blasting
Central (Hall 2, Hall 3)	8–18	Pad footings on till where cover to rock exceeds ≈ 6 m, transitioning to rock-founded piers where cover is shallower, confirmed hole-by-hole
East (Hall 4, generator compound)	19–24	Direct-to-rock or rock-socketed piers; controlled blasting required

4.3 Blasting. Controlled blasting in the eastern third shall be designed by a licensed blasting contractor, with scaled-distance calculations and a vibration monitoring regime complying with SFS 4812 peak particle velocity (PPV) guide values at the nearest sensitive receptors (the north-east residential boundary and the Tikkurilantie logistics neighbour). Pre-blast condition surveys of these structures shall be completed and a louhintailmoitus

submitted to the Vantaa building control and environmental authorities, with written notice to affected neighbours, before any blast is fired.

4.4 Boulder-affected zone. Between BH-07 and BH-09 (PDC-SS-001 §7a), the Contractor shall allow for reduced excavation and piling productivity and shall not finalise piling methodology in this zone without the targeted supplementary investigation recommended in PDC-SS-001.

4.5 Sulphide-affected material. Material excavated within approximately 15 m of BH-12 between 3.2 m and 4.6 m depth shall be segregated, stockpiled separately on an impermeable base, and not mixed with general till arisings pending waste classification (PDC-SS-001 §7c).

4.6 Dewatering. Excavations extending below approximately 2.5 m shall anticipate perched groundwater at 2.6–3.9 m depth (seasonal variation 0.5–1.0 m) and shall be dewatered by sump pumping or well-pointing as a minimum, with higher allowances in the boulder-rich horizon at BH-07/BH-08/BH-09.

4.7 Ground-bearing slab. The slab shall be placed on a proof-rolled, compacted granular capping layer over a damp-proof membrane, designed for a 25 kN/m² uniform imposed load throughout the white space and support areas, to FM2 flatness class. No raised access floor is provided in the white space; overhead containment and busway carry all cabling (refer PDC-SPEC-04).

4.8 Curing and jointing. Concrete shall be cured in accordance with SFS-EN 13670; slab joints, saw-cut or formed, shall be located to coordinate with the primary structural grid and shall not be positioned under transformer, generator, UPS or CRAH plinths.

5. Tolerances

Item	Tolerance
Formation level, general excavation	± 50 mm
Formation level, blasted rock surface	+ 150 mm / – 0 mm (overbreak to be made good)
Pad/strip footing plan position	± 25 mm
Pier/pile plan position	± 50 mm
Pier/pile verticality	1:75
Top-of-footing level	± 15 mm
Ground-bearing slab flatness	FM2 per SFS-EN 13670 (25 mm over 3 m straightedge, general areas)
Ground-bearing slab level	± 10 mm from datum
Damp-proof membrane lap	Minimum 300 mm, taped

6. Testing and Inspection

Test / inspection	Frequency / trigger	Acceptance basis
Proof-rolling of formation	Every foundation and slab area before blinding/capping	No visible deflection or rutting under specified plant
Plate load test	Minimum one per foundation zone (west/central/east)	Bearing pressures per PDC-SS-001 §6.2 table, confirmed at formation level
Pile/pier integrity test (where used)	100% of piers in boulder-affected and rock-socketed zones	No indication of debonding, necking or damage
Concrete cube testing	Per SFS-EN 206 sampling frequency, per pour	28-day characteristic strength achieved for the specified class

Vibration monitoring during blasting	Continuous at nearest sensitive receptors during blasting works	SFS 4812 PPV guide values not exceeded
Groundwater/piezometer monitoring	Monthly, standpipes at BH-02, BH-06, BH-10, BH-12	Confirms dewatering design assumptions
Sulphide/waste classification testing	Segregated stockpile from BH-12 zone, before offsite disposal	Waste classification confirmed before removal from site
Slab flatness survey	100% of white space and support-area slab, post-cure	FM2 class achieved

7. Submittals

Submission	Timing
Geotechnical design report (Eurocode 7 Design Approach used, bearing values by zone)	S2 detailed design
Blasting method statement, scaled-distance calculations, vibration monitoring plan	Minimum 21 days before blasting commences
Pre-blast condition surveys of the north-east residential boundary and the logistics neighbour	Before blasting commences
Dewatering method statement	S2 detailed design, updated at S3
Concrete mix designs for all classes listed in Section 3	S3 construction issue
Foundation setting-out and as-built survey	On completion of each foundation zone
Waste classification and disposal routing plan for sulphide-affected material	Before bulk excavation reaches the BH-12 zone
Slab flatness survey report	On completion of each hall's slab

8. Warranties

8.1 The Contractor shall warrant the ground-bearing slab against defects in flatness, cracking and curling for a minimum period of 12 months from the relevant Section's Taking-Over, without prejudice to any longer statutory liability period under Finnish law.

8.2 Waterproofing (damp-proof membrane) shall carry the manufacturer's standard material warranty, submitted with the S4 as-built documentation for each Section.

End of PDC-SPEC-01 Rev A.